Improving I/O Performances of HPC Application Gysela

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Abstract: In modern High-Performance Computing (HPC) systems, Input/Output (I/O) performance is a major bottleneck for large-scale scientific simulations. As computational power and data volumes scale rapidly, optimizing I/O workflows becomes critical to efficiently leverage supercomputing resources. Scientific applications heavily rely on Parallel File Systems (PFS) to access vast datasets concurrently. However, achieved performance is highly sensitive to PFS configurations, high-level I/O strategies, and application-specific access patterns. Understanding and tuning the interactions between these layers is crucial for maximizing global simulation throughput.
This work focuses on optimizing the checkpointing performance of Gysela, a 5D gyrokinetic simulation code developed by the CEA for nuclear fusion research. Gysela generates massive datasets and relies on periodic checkpointing for fault tolerance and job restartability. Using gys_io, a lightweight mini-app replicating Gysela's parallel I/O patterns, we systematically evaluate and compare different parallel I/O strategies—namely Single Shared File (SSF), Subfiling, and File Per Process (FPP)—alongside key PFS configuration parameters. Our experiments, conducted on diverses clusters, analyze the performance trade-offs of these strategies and parameters under various scaling conditions. The results identify optimal parallel configurations, clarify the impact of PFS striping on each I/O strategy, and provide actionable optimization recommendations to significantly reduce checkpointing overhead in future Gysela production runs.

Key-words: HPC, Parallel I/O, Performances Tuning, Parallel File Systems

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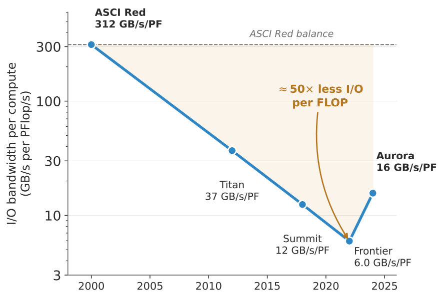
Contents

1 Introduction

- I/O is almost important as computation in HPC.
- As supercomputer advance, computational power increases faster thant storage bandwidth, making I/O operation a bottleneck.
  - Find an exemple as Luan's example with ratio between computation and I/O perfs
  - `https://arxiv.org/pdf/2501.00203` <- article about difference of treatment between computation and I/O in HPC

  ratio between computation TFlops and I/O GB/s for top 500 supercomputer from 2020 to 2025.
- Large-Scale simulations generate unprecedented volumes of datas, much be stored and retrieved efficiently.
  - Example like dataset for AI training (ChatGPT), weather or any kind of simulation
- I/O in parallel file system is a complex problem due to concurrent access, contention, and data distribution.
- Gysela is a 5D gyrokinetic simulation code developed by CEA for nuclear fusion research (plasma turbulence in tokamaks).
- Run on large amount of nodes and generate massives datasets
  - Example of dataset size for Gysela
  - -> Ask Virginie about diverses size for Gysela prod
- Due to long execution time and risk of hardware failure, Gysela relies on periodic (every X [ask virginie] iterations) checkpointing to save its state and allow clean restart
- Writing checkpoint is synchronus and blocks the simulation progress, wasting expensive resources.
- On production, checkpoint write time suffer from high variability sometimes up to x3 slower than average.
- We must optimize both the PFS configuration and the I/O strategy
- We use gys_io, a lightweight mini-application that replicates Gysela's 5D grid checkpointing pattern.

  - Explain more how gys_io works

- We leverage PDI (Parallel Data Interface) to implement different I/O strategies: Single Shared File (SSF), Subfiling, and File Per Process (FPP) by modifing the configuration file in yaml, for flexible execution and neither code modification and compilation.
- We submits tests and evaluate the performances with IOPS, a benchmarking tool developped by TADaaM team at Inria Bordeaux.
- Our eval is conducted on different cluster with different PFS : Plafrim (with BeegFS) and Adastra (with Lustre). This allow us to test different PFS configuration and analyze the impact of striping on each I/O strategy.
- This work aims to identify optimal configurations and provide actionable recommendations to reduce checkpointing overhead in future Gysela production runs.

2 Background

Text for the second section. This is closely related to the text in section 1 on page 4.

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